



**WORKING CAPITAL EFFICIENCY: A COMPARATIVE STUDY OF TWO CYCLE
SHOPS IN SYLHET**

Course: Working Capital Management

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Introduction

This assignment is based on independent field visits to two cycle-related retail shops in Sylhet. Using a structured interview guide, primary data was collected on inventory holding period, supplier payment terms, customer credit practices, and monthly sales volume. Based on the collected data, individual Excel models were developed to calculate key working capital metrics including Operating Working Capital (OWC), Working Capital Requirement (WCR), Cash Conversion Cycle (CCC), and WCR/Sales ratio. All calculations were derived using a consistent daily-sales-based approach to ensure full alignment between the written analysis and the Excel model. After completing the models, only the final CCC and WCR/Sales ratios were shared between peers to conduct a comparative analysis. The objective of this assignment is to identify operational differences that influence working capital efficiency and to provide specific, actionable recommendations based on real business observations.

Business Description - Shop A: Sylhet Cyclewala & Auto Parts

Sylhet Cyclewala & Auto Parts is a small-scale cycle gadget and repair shop located in the Bateshwar Asha Bank area of Sylhet. The business is owned and operated by Mr. Masud Rana. The shop primarily sells bicycle parts such as handlebars, cassettes, and repair accessories. The shop operates within a limited physical space with no separate storage facility. Inventory is stored within the shop premises, indicating a reliance on frequent restocking rather than bulk purchasing. During the visit, the shop appeared compact but organized, with inventory visibly displayed rather than stored in bulk. The owner showed a strong preference for cash transactions and responded cautiously when discussing credit sales, suggesting a deliberate attempt to avoid receivables risk. However, the reliance on cash-on-delivery purchasing indicates limited bargaining power with suppliers, which directly impacts the shop's liquidity position.

Key Interview Data - Shop A (taken by Dhrubo)

Parameter	Detail
Monthly Sales	BDT 100,000
Inventory Holding (DIO)	30 days
Supplier Payment Terms (DPO)	0 days (Cash on Delivery)
Customer Credit Period (DSO)	30 days
Credit vs Cash Split	97% Cash / 3% Credit

Working Capital Model - Shop A

All calculations are derived using daily sales to maintain consistency with the Excel model.

$$\text{Daily Sales} = 100,000 / 30 = 3,333.33$$

$$\text{Accounts Receivable (AR)} = (3,333.33 \times 3\%) \times 30 = \text{BDT } 3,000$$

$$\text{Inventory} = 3,333.33 \times 30 = \text{BDT } 100,000$$

$$\text{Accounts Payable (AP)} = 3,333.33 \times 0 = \text{BDT } 0$$

Working Capital Metrics - Shop A

Metric	Value
Cash Conversion Cycle (CCC)	60 days
Operating Working Capital (OWC)	BDT 103,000
Working Capital Requirement (WCR)	BDT 103,000
WCR / Sales Ratio	1.03

The shop experiences a 60-day cash conversion cycle, primarily due to the mismatch between DSO (30 days) and DPO (0 days). This means the owner must pay suppliers immediately while waiting up to a month to collect cash from customers, creating a significant liquidity gap.

Business Description - Shop B: Jonota Cycle Gallery

Jonota Cycle Gallery is a cycle retail business located in the ShahPoran area of Sylhet, owned by Mr. Jomir Ahmed. The shop sells bicycles, tricycles, and related parts, offering a broader product range compared to Shop A. The business benefits from strong supplier relationships, allowing delayed payment of dues. Inventory turnover is relatively faster, and customer credit periods are shorter, contributing to a more efficient working capital cycle.

Key Interview Data - Shop B (taken by Aticullah)

Parameter	Detail
Monthly Sales	BDT 90,000
Inventory Holding (DIO)	15 days
Supplier Payment Terms (DPO)	30 days (based on flexible settlement terms)
Customer Credit Period (DSO)	5 days

Parameter	Detail
Credit vs Cash Split	95% Cash / 5% Credit

Working Capital Model - Shop B

All calculations are derived using daily sales to maintain consistency with the Excel model.

$$\text{Daily Sales} = 90,000 / 30 = 3,000$$

$$\text{Accounts Receivable (AR)} = (3,000 \times 5\%) \times 5 = \text{BDT } 750$$

$$\text{Inventory} = 3,000 \times 15 = \text{BDT } 45,000$$

$$\text{Accounts Payable (AP)} = 3,000 \times 30 = \text{BDT } 90,000$$

Working Capital Metrics - Shop B

Metric	Value
Cash Conversion Cycle (CCC)	-10 days
Operating Working Capital (OWC)	-BDT 44,250
Working Capital Requirement (WCR)	-BDT 44,250
WCR / Sales Ratio	-0.49

A negative CCC indicates that the shop collects cash from customers before paying suppliers, allowing it to operate using supplier financing and reducing liquidity pressure.

Comparative Analysis of Working Capital Performance

Metric	Shop A	Shop B
CCC	60 days	-10 days
WCR/Sales	1.03	-0.49

Shop A's higher CCC reflects delayed cash recovery and immediate supplier payments, while Shop B's negative CCC demonstrates efficient cash flow management supported by supplier credit and faster inventory turnover.

Peer Comparison of CCC and WCR/Sales Ratio

My shop's (Shop – A) CCC is 60 days, whereas my pair's shop (Shop – B) has a CCC of -10 days. Similarly, my shop's WCR/Sales ratio is 1.03, while my pair's shop operates at -0.49, indicating a significantly stronger working capital position. Two specific reasons explain this difference. First, my shop operates on a cash-on-delivery basis (DPO = 0 days), requiring immediate payment to suppliers, while my pair's shop benefits from extended supplier credit (approximately 30 days) due to stronger supplier relationships. Second, my shop holds inventory for a longer period (30 days), whereas my pair's shop maintains faster turnover (15 days), reducing capital lock-up. These operational differences result in significantly higher cash pressure in my shop.

Which Shop Has the Stronger Working Capital Position?

Jonota Cycle Gallery (Shop B) has the stronger working capital position. This is because it operates with a negative CCC (-10 days) and a negative WCR/Sales ratio (-0.49), meaning it collects cash before paying suppliers. In contrast, Shop A's positive CCC and WCR indicate significant capital is tied up in operations, resulting in weaker liquidity.

What-If Scenario Analysis (Shop A)

Scenario: Customer payment delay increases by 14 days

New DSO = $30 + 14 = 44$ days
Daily Sales = 3,333.33
Daily Credit Sales = $3,333.33 \times 3\% = 100$
New AR = $100 \times 44 = \text{BDT } 4,400$
Inventory (unchanged) = BDT 100,000
Accounts Payable (unchanged) = BDT 0

Metric	Before	After
CCC	60 days	74 days
OWC	103,000	104,400
WCR/Sales	1.03	1.04

Interpretation: A 14-day increase in customer payment period significantly worsens Shop A's liquidity position. The CCC increases from 60 to 74 days, further delaying cash recovery, while working capital requirements rise due to higher receivables. Since the shop already operates without supplier credit, this additional delay directly increases dependence on internal funds, intensifying cash flow pressure.

What-If Scenario Analysis (Shop B)

Scenario: Customer payment delay increases by 14 days

New DSO = $5 + 14 = 19$ days
Daily Sales = 3,000
Daily Credit Sales = $3,000 \times 5\% = 150$
New AR = $150 \times 19 = \text{BDT } 2,850$
Inventory (unchanged) = BDT 45,000
Accounts Payable (unchanged) = $3,000 \times 30 = \text{BDT } 90,000$

Metric	Before	After
CCC	-10 days	4 days

Metric	Before	After
OWC	-44,250	-42,150
WCR/Sales	-0.49	-0.47

Interpretation: A 14-day increase in customer payment period significantly weakens Shop B's working capital efficiency. The CCC shifts from -10 days to +4 days, meaning the business no longer benefits from a negative cash cycle and must now wait to recover cash after paying suppliers. Although the shop still operates with negative working capital, the reduction in supplier-financed advantage increases liquidity pressure and reduces financial flexibility.

Key Findings

- The mismatch between DSO and DPO is the primary cause of liquidity pressure in Shop A.
- Even small amounts of credit sales can significantly affect cash flow when collection periods are long.
- Inventory turnover plays a critical role in working capital efficiency.
- Strong supplier relationships can enable businesses to operate with negative working capital, improving financial flexibility.

Recommendation for Shop A

The main issue in Shop A is the mismatch between DSO (30 days) and DPO (0 days), which increases the Cash Conversion Cycle (CCC) and creates a financing gap. The owner should immediately negotiate 7-10 days of trade credit from suppliers. This will directly reduce the CCC and improve liquidity by delaying cash outflows.

Recommendation for Shop B

Shop B already demonstrates strong working capital efficiency. However, further improvement can be achieved by reducing slow-moving inventory and aligning purchases more closely with demand patterns.

Conclusion

This assignment highlights how operational decisions directly influence working capital efficiency. Despite operating in the same industry, the two shops exhibit significantly different financial positions due to differences in supplier credit, inventory turnover, and receivable management. Shop A faces liquidity pressure due to immediate supplier payments and delayed cash recovery, whereas Shop B benefits from efficient operations and favorable credit terms, allowing it to maintain a negative cash conversion cycle. Effective management of DIO, DSO, and DPO is therefore essential for improving financial stability and sustaining business operations.

Appendix A

Interview Answer Details – Dhrubo Barua Dip (Sylhet Cyclewala & Auto Parts, Shop A)

FIN4707 | Interview Guide | My Shop's Working Capital | AIBA Sylhet

WCM Assignment— Interview Guide

MyShop'sWorkingCapital

Bring this sheet to the shop. Fill it in during or immediately after your visit. Handwriting is fine.

SECTION A — Your Details & Shop Identification

Your name & student ID	Dhrubo Barua Dip - 101022027
Your pair's name & student ID	Aticullah Khan - 101022060
industry (e.g. grocery, pharmacy, hardware)	Cycle Gadget & Repair Shop
Shop / business name	Sylhet Cyclewala & Auto Parts
Owner or manager name & Contact info.	Masud Rana - 01817110066
Address — area, road, or market name	Bateshwar Asha Bank

SECTION B — Inventory (feeds DIO calculation)

Ask: 'How long does your stock typically sit before it is sold?'

	Question to ask	Their answer — write exactly what they say
B1	What is your main product or product category? (be specific — 'rice and lentils', not just 'groceries')	Bicycle Parts
B2	How many days does stock typically sit before it is sold? Or: how often do you reorder?	Approx. 1 Month
B3	Is there any product that sits much longer than usual? How long and why?	Bicycle Handlebar, Cassette
B4	Do you ever run out of stock? What happens when you do?	Till now not

SECTION C — Supplier Credit (feeds DPO calculation)

Ask: 'When you buy from your supplier, when do you actually pay them?'

	Question to ask	Their answer
C1	Who is your main supplier? (name, location, or description — e.g. 'a Chattogram importer')	Sadarghat Road, Chattogram - Bicycle wholesale market
C2	How many days do you have to pay after an order? Or is it cash on delivery?	Cash on Delivery
C3	Has this arrangement changed recently? Did they tighten or loosen terms, and why?	No. Sometimes terms get loosened if you are a regular & loyal buyer.
C4	Do you get any discount for paying early? Have you ever taken it?	Get some discounts, as it's a wholesale market.

SECTION D — Customer Credit (feeds DSO calculation)

Ask: 'Do you sell to any customers on credit, or is everything cash?'

	Question to ask	Their answer
D1	Do any customers buy on credit? If yes, who — regular customers, other businesses, institutions?	Yes. Mostly familiar customers
D2	How many days do they typically take to pay? What is the longest you have waited?	Within 30 days
D3	What percentage of your sales are on credit vs. cash? A rough guess is fine.	97% Cash, 3% Credit
D4	Have you ever had a customer not pay? What did you do?	No

SECTION E — Cash Position & Sales Volume

These numbers feed your WCR/Sales ratio and what-if scenario.

	Question to ask	Their answer
E1	Roughly what are your monthly sales? A range is fine — e.g. 'BDT 1–2 lakh per month'.	Approx. BDT 1 lakh per month
E2	Do any customers pay you in advance before receiving goods?	No
E3	Do you pay any regular bills in advance — rent, insurance, or licence fees?	No
E4	Is there a season — Eid, harvest, exam time — when cash is tightest? What happens then?	No
E5	Has cash ever been tight enough that you could not pay a supplier on time? What caused it?	No

Your Observer Notes

Write anything you noticed that the owner did not say — the stockroom, how they reacted to questions about credit, visible signs of cash stress, anything that adds context to the numbers.

- During the visit, the shop appeared small but well-organized, with most bicycle parts visibly displayed in sufficient quantities.
- There was no dedicated storage system or warehouse; inventory seemed to be kept within the shop space itself. This indicates limited working capital and a reliance on frequent restocking rather than on bulk purchasing.
- The owner showed a clear preference for cash transactions. When discussing credit sales, he responded briefly, reinforcing that credit is minimal (around 3%) and likely restricted to trusted, familiar customers only. He follows a deliberate strategy to avoid receivables risk.
- On the supplier side, the mention of "cash on delivery" aligned with the nature of the wholesale market. However, the owner's tone implied a lack of bargaining power.
- In times of cash shortage, the owner depends on personal networks instead of formal financing, highlighting limited access to institutional credit. Overall, the business operates with tight liquidity and minimal margin for financial disruption.

Tips for a Good Visit

Introduce yourself clearly: 'I am a student from AIBA doing a class project on how small businesses manage cash.' Most owners are glad to help students.

Ask indirectly about money: Instead of 'what is your monthly revenue?', try 'how many customers do you get on a typical day?' Then estimate together.

Let them go off-script: The best data often comes after the direct question. 'And during Eid...' is often more revealing than the official answer.

Fill this in immediately: Write up your answers while you are still outside the shop — not the next morning.

Messy numbers are good: If all your answers are round numbers, a grader will notice. Real conversations produce specific, slightly irregular figures.

Assure confidentiality: Tell the owner that specific figures will only be seen by your instructor for academic analysis.

Appendix B

Interview Answer Details – Aticullah Khan (Jonota Cycle Gallery, Shop B)

WCM Assignment — Interview Guide
Bring this sheet to the shop. Fill it in during or immediately after your visit. Handwriting is fine.

SECTION A — Your Details & Shop Identification

Your name & student ID	Aticullah Khan(101022060)
Your pair's name & student ID	Dhrubo Barua Dip - 101022027
industry (e.g. grocery, pharmacy, hardware)	Cycle Shop
Shop / business name	Jonota Cycle Gallery
Owner or manager name & Contact info.	Jomir Ahmed(01758766797)
Address — area, road, or market name	M R A Complex, Sha Poran

SECTION B — Inventory (feeds DIO calculation)
Ask: 'How long does your stock typically sit before it is sold?'

	Question to ask	Their answer — write exactly what they say
B1	What is your main product or product category? (be specific — 'rice and lentils', not just 'groceries')	Bycycle, Tri-cycle and Parts
B2	How many days does stock typically sit before it is sold? Or: how often do you reorder?	15 days
B3	Is there any product that sits much longer than usual? How long and why?	Bycycle. Because of limited options
B4	Do you ever run out of stock? What happens when you do?	Yes, Tri-cycle and parts. Ordered them on demand

SECTION C — Supplier Credit (feeds DPO calculation)
Ask: 'When you buy from your supplier, when do you actually pay them?'

	Question to ask	Their answer
C1	Who is your main supplier? (name, location, or description — e.g. 'a Chattogram importer')	Distributors of Sylhet district
C2	How many days do you have to pay after an order? Or is it cash on delivery?	There is no fixed date. But I have to pay all the remaining dues in December of that year.
C3	Has this arrangement changed recently? Did they tighten or loosen terms, and why?	No. Because of my good relation the terms get loosen.
C4	Do you get any discount for paying early? Have you ever taken it?	I don't get discount on paying early but I get on quantities of order

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Ask: 'Do you sell to any customers on credit, or is everything cash?'

	Question to ask	Their answer
D1	Do any customers buy on credit? If yes, who — regular customers, other businesses, institutions?	Yes. Regular and familiar customers
D2	How many days do they typically take to pay? What is the longest you have waited?	4-5 days
D3	What percentage of your sales are on credit vs. cash? A rough guess is fine.	95% cash and 5% credit
D4	Have you ever had a customer not pay? What did you do?	No

SECTION E—Cash Position & Sales Volume

These numbers feed your WCR/Sales ratio and what-if scenario.

	Question to ask	Their answer
E1	Roughly what are your monthly sales? A range is fine — e.g. 'BDT 1–2 lakh per month'.	80000-100000tk
E2	Do any customers pay you in advance before receiving goods?	Yes
E3	Do you pay any regular bills in advance — rent, insurance, or licence fees?	No
E4	Is there a season — Eid, harvest, exam time — when cash is tightest? What happens then?	Yes, I borrow money from my relative.
E5	Has cash ever been tight enough that you could not pay a supplier on time? What caused it?	No

Your Observer Notes

Write anything you noticed that the owner did not say — the stockroom, how they reacted to questions about credit, visible signs of cash stress, anything that adds context to the numbers.

- During the visit, the shop appeared well-organized and professionally maintained, with clearly separated sections for different types of bicycles. Unlike many small businesses, the layout suggested structured inventory management and customer convenience.
- The owner demonstrated strong business confidence and experience. He mentioned that he could even open another shop without personal investment, relying entirely on supplier credit, which reflects high trust and strong relationships with distributors.
- Inventory management seemed efficient, with a low holding period and quick replenishment strategy. Instead of overstocking, the owner depends on demand-based ordering, reducing holding costs and risk.
- An interesting observation was the informal recycling system. The owner purchases old bicycles and reuses parts for repair and resale. This indicates cost efficiency, resource optimization, and additional profit generation.
- The business maintains a very strong cash position, with minimal reliance on customer credit (only around 5%). This reduces the risk of bad debts and ensures steady liquidity.
- Overall, the business operates with stable liquidity, strong supplier relationships, efficient inventory control, and low financial risk, making it a well-managed small-scale retail operation.

Tips for a Good Visit

Introduce yourself clearly: 'I am a student from

AIBA doing a class project on how small businesses manage cash.' Most owners are glad to help students.

Ask indirectly about money: Instead of 'what is your monthly revenue?', try 'how many customers do you get on a typical day?' Then estimate together.

Let them go off-script: The best data often comes after the direct question. 'And during Eid...' is often more revealing than the official answer.

Fill this in immediately: Write up your answers while you are still outside the shop — not the next morning.

Messy numbers are good: If all your answers are round numbers, a grader will notice. Real conversations produce specific, slightly irregular figures.

Assure confidentiality: Tell the owner that specific figures will only be seen by your instructor for academic analysis.

Appendix C

Input Tab - Dhrubo (Shop - A)

Parameter	Detail
Monthly Sales	BDT 100,000
Inventory Holding (DIO)	30 days
Supplier Payment Terms (DPO)	0 days (Cash on Delivery)
Customer Credit Period (DSO)	30 days
% of Cash Sales	97%
% of Credit Sales	3%
Accounts Payable	BDT 0
Accounts Receivable	BDT 3,000
Inventory	BDT 100,000

Daily Sales = 100,000 / 30 = 3,333.33

Accounts Receivable (AR) = (3,333.33 × 3%) × 30 = BDT 3,000

Inventory = 3,333.33 × 30 = BDT 100,000

Accounts Payable (AP) = 3,333.33 × 0 = BDT 0

Model Tab - Dhrubo (Shop - A)

Metric	Value
Cash Conversion Cycle (CCC)	60 days
Operating Working Capital (OWC)	BDT 103,000
Working Capital Requirement (WCR)	BDT 103,000
WCR / Sales Ratio	1.03

What-If Tab - Dhrubo (Shop - A)

Scenario: Customer payment delay increases by 14 days

New DSO = 30 + 14 = 44 days

Daily Sales = 3,333.33

Daily Credit Sales = 3,333.33 × 3% = 100

New AR = 100 × 44 = BDT 4,400

Inventory (unchanged) = BDT 100,000

Accounts Payable (unchanged) = BDT 0

Metric	Before	After
Cash Conversion Cycle (CCC)	60 days	74 days
Operating Working Capital (OWC)	103,000	104,400
Working Capital Requirement (WCR/Sales) Ratio	1.03	1.04

Interpretation: A 14-day increase in customer payment period significantly worsens Shop A's liquidity position. The CCC increases from 60 to 74 days, further delaying cash recovery, while working capital requirements rise due to higher receivables. Since the shop already operates without supplier credit, this additional delay directly increases dependence on internal funds, intensifying cash flow pressure.

Appendix D

Input Tab - Aticullah (Shop - B)					
Parameter	Detail				
Monthly Sales	BDT 90,000				
Inventory Holding (DIO)	15 days	Daily Sales = 90,000 / 30 = 3,000			
Supplier Payment Terms (DPO)	30 days (based on flexible settlement terms)				
Customer Credit Period (DSO)	5 days	Accounts Receivable (AR) = (3,000 × 5%) × 5 = BDT 750			
% of Cash Sales	95%				
% of Credit Sales	5%	Inventory = 3,000 × 15 = BDT 45,000			
Accounts Payable	BDT 90,000				
Accounts Receivable	BDT 750	Accounts Payable (AP) = 3,000 × 30 = BDT 90,000			
Inventory	BDT 45,000				

Model Tab - Aticullah (Shop - B)	
Metric	Value
Cash Conversion Cycle (CCC)	-10 days
Operating Working Capital (OWC)	-BDT 44,250
Working Capital Requirement (WCR)	-BDT 44,250
WCR / Sales Ratio	-0.49

What-If Tab - Aticullah (Shop - B)		
Scenario: Customer payment delay increases by 14 days		
New DSO = 5 + 14 = 19 days		
Daily Sales = 3,000		
Daily Credit Sales = 3,000 × 5% = 150		
New AR = 150 × 19 = BDT 2,850		
Inventory (unchanged) = BDT 45,000		
Accounts Payable (unchanged) = 3,000 × 30 = BDT 90,000		
Metric	Before	After
Cash Conversion Cycle (CCC)	-10 days	4 days
Operating Working Capital (OWC)	-44,250	-42,150
Working Capital Requirement (WCR/Sales) Ratio	-0.49	-0.47
Interpretation: A 14-day increase in customer payment period significantly weakens Shop B's working capital efficiency. The CCC shifts from -10 days to +4 days, meaning the business no longer benefits from a negative cash cycle and must now wait to recover cash after paying suppliers. Although the shop still operates with negative working capital, the reduction in supplier-financed advantage increases liquidity pressure and reduces financial flexibility.		